

Project Executable Files

Date	08 July 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	3 Marks

Project Executable Files

This section requires submission of all executable and deployable files for your project. Ensure all files are properly organized, named, and uploaded to the specified submission platform. The evaluator must be able to run or deploy your solution using the provided files and instructions.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	NA — no database used
5	Environment configuration file (.env.example or similar)	Yes
6	Deployed application URL (if hosted)	Yes
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA — web app, no binary
8	Dockerfile / Containerization config (if applicable)	No — not containerized
9	Test files and test results	Yes — 8/8 tests passing
10	Demo video or walkthrough (if required)	NA

Step 2: File / Folder Structure

Overview of the project's file and folder structure:

```
hdi-prediction-system/
├── app.py                — Streamlit application entry point
├── hdi_utils.py          — UNDP HDI formula implementation
├── train_model.py        — Generates training data & trains ML models
├── requirements.txt      — Python dependencies
├── .env.example          — Environment configuration template
├── .gitignore
├── models/              — Trained model artifacts
│   ├── hdi_regressor.joblib
│   ├── hdi_classifier.joblib
│   └── feature_columns.joblib
├── tests/
│   └── test_hdi_utils.py — Unit tests for the HDI formula
└── README.md
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	https://hdi-prediction-system-aetdgnddijavzfycsxmi9n.streamlit.app/
Demo Video link	https://drive.google.com/file/d/1RU7i5S6H5etNvUD8rkscL35g8hGt7mwF/view?usp=drivesdk
Platform / Hosting Provider	Streamlit
Repository Link	https://github.com/vikashchitikela/A-Comprehensive-Measure-of-Well-Being

Step 4: Run Instructions

Pre-requisites:**Pre-requisites:**
Python 3.10+ and pip installed.

Step 1: Extract/clone the project files

Step 2: (Recommended) Create a virtual environment: `python3 -m venv venv && source venv/bin/activate`

Step 3: Install dependencies: `pip install -r requirements.txt`

Step 4: Configure environment: `cp .env.example .env`

Step 5: Train the models (already included, optional): `python train_model.py`

Step 6: Start the application: `streamlit run app.py`

Step 7: Open browser at `http://localhost:8501`

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	ML models are trained on synthetically generated data consistent with the UNDP formula, not real historical country HDI datasets	Acceptable for demonstration; retrain on official UNDP Human Development Report data for production use
2	No persistent database — each session is stateless	By design; not required for current scope